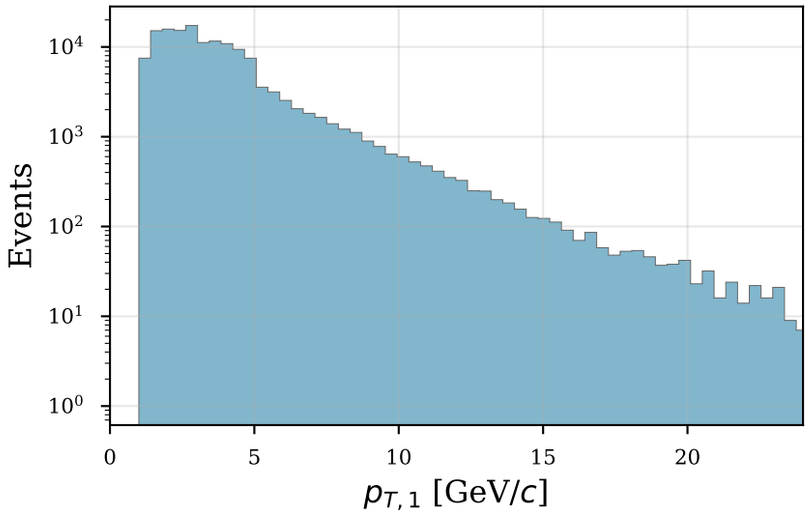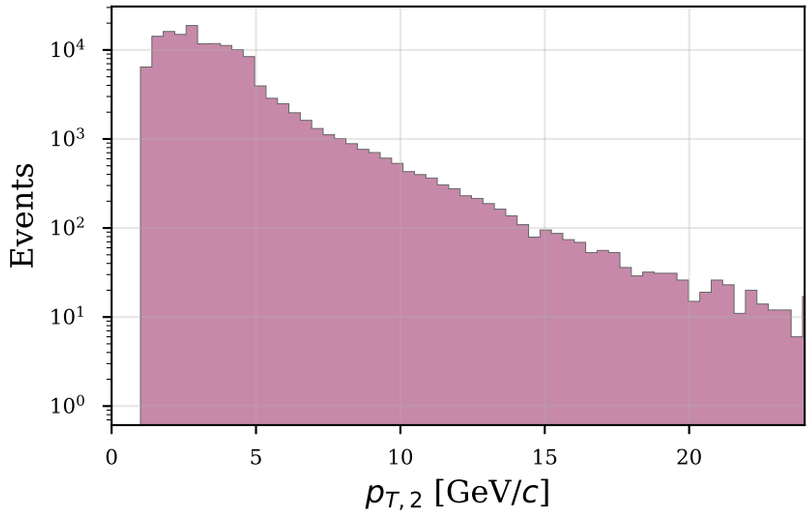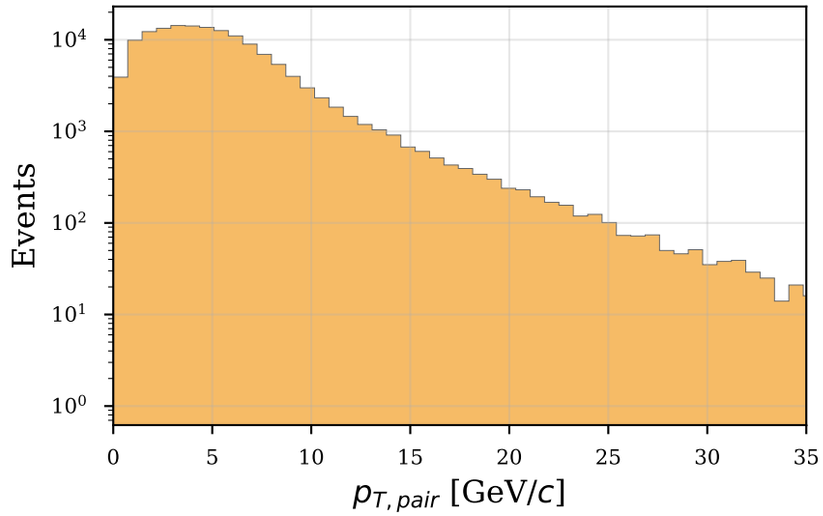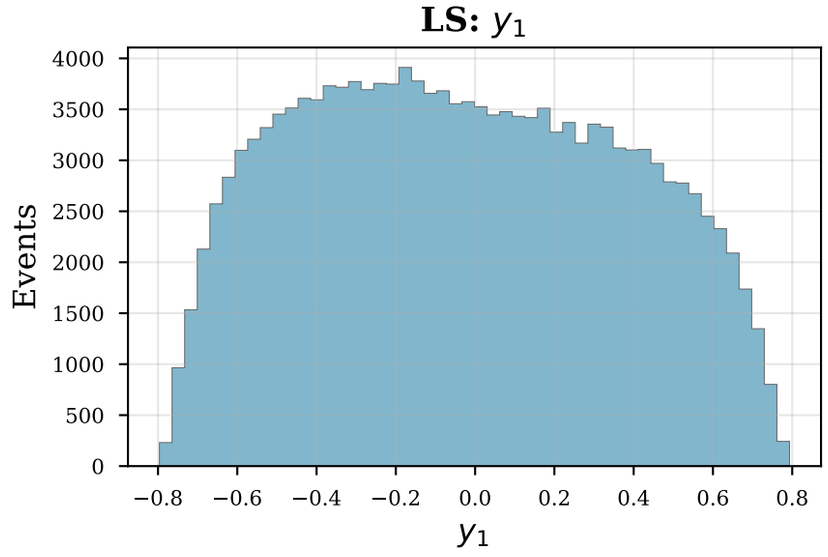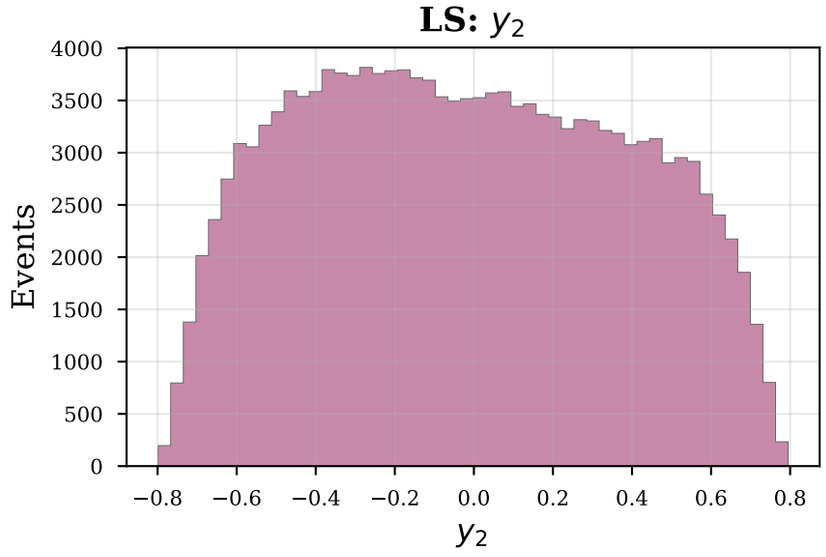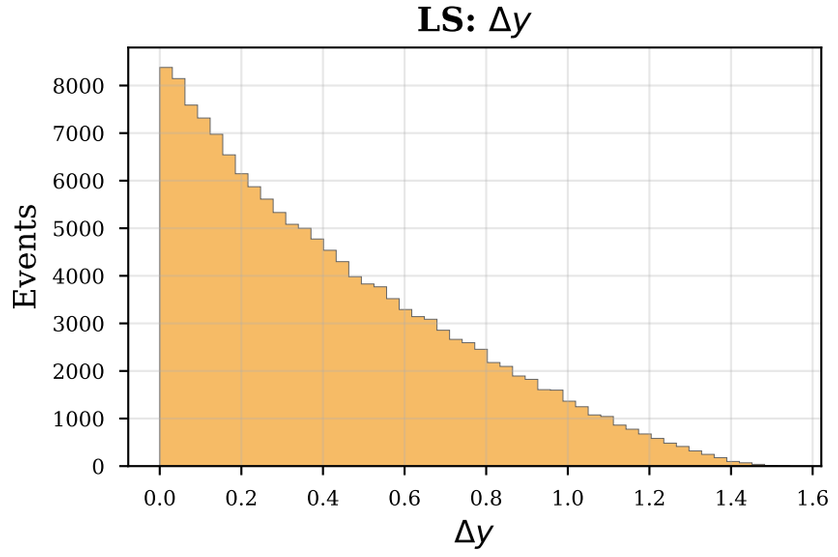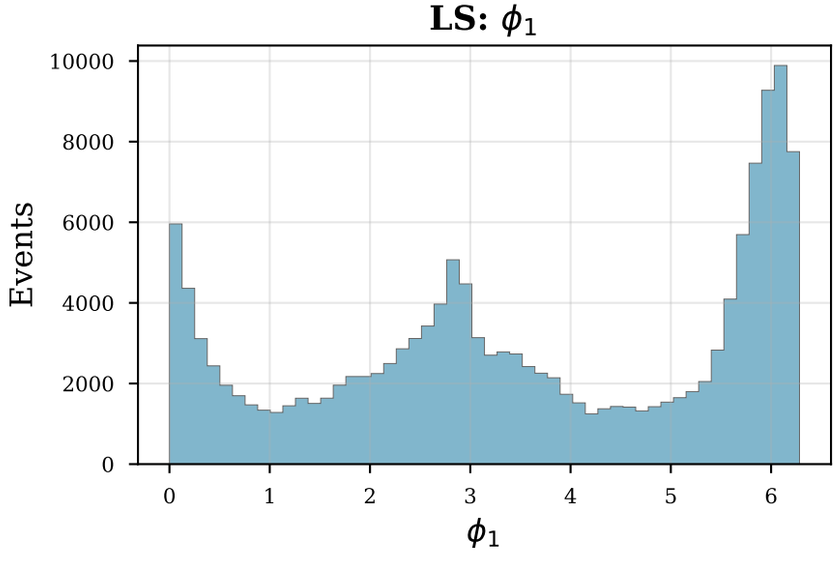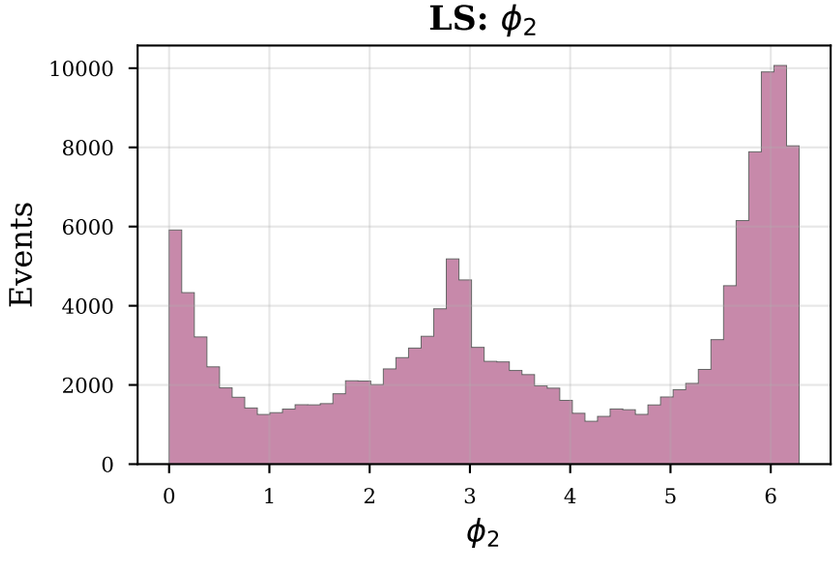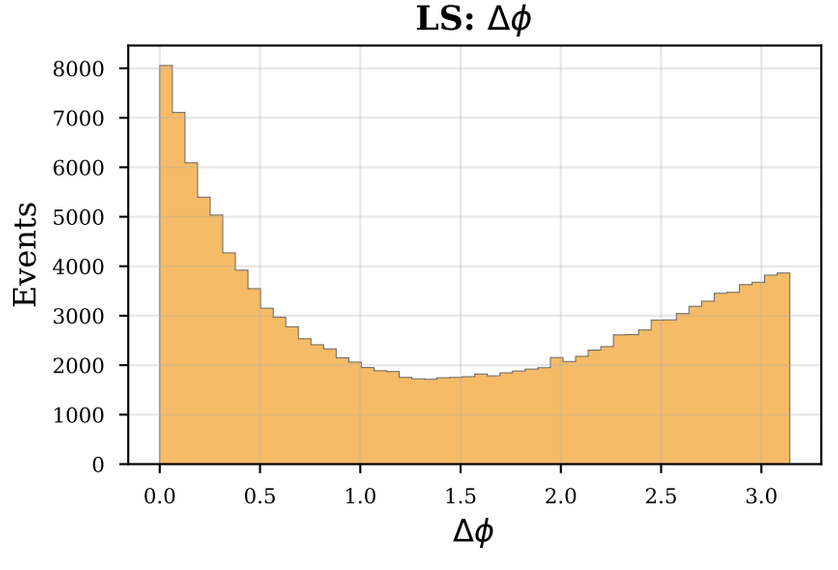LS Kinematic Distributions